J2EE & Python Practical Programs

Copy-paste ready code with setup steps, execution instructions, and sample outputs.

Practical 1: Insert Student Details using JSP, Servlet and JDBC

Aim: Insert Student ID, Name, Course and City into a MySQL database.

Step 1 - Create database and table

Open MySQL Workbench or phpMyAdmin and execute:

```
CREATE DATABASE college;
USE college;
CREATE TABLE student (
    stud_id INT PRIMARY KEY,
    stud_name VARCHAR(50),
    stud_course VARCHAR(50),
    stud_city VARCHAR(50)
);
```

Step 2 - Create Eclipse Dynamic Web Project

File > New > Dynamic Web Project. Name: **StudentRegistration**. Select **Apache Tomcat 9** as target runtime. Add the MySQL Connector/J JAR to **WEB-INF/lib**.

Step 3 - Create index.jsp

Create **index.jsp** under WebContent (or src/main/webapp). Paste:

```
<!DOCTYPE html>
<html>
<head><title>Student Registration</title></head>
<body>
<h2>Student Registration Form</h2>
<form action="StudentServlet" method="post">
    Student ID: <input type="number" name="id" required><br><br>
    Student Name: <input type="text" name="name" required><br><br>
    Student Course: <input type="text" name="course" required><br><br>
    Student City: <input type="text" name="city" required><br><br>
    <input type="submit" value="Insert Student">
</form>
</body>
</html>
```

Step 4 - Create StudentServlet.java

Create a Servlet in package **com.student** named **StudentServlet**. Replace its code with:

```
package com.student;

import java.io.IOException;
import java.io.PrintWriter;
import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.PreparedStatement;
import javax.servlet.ServletException;
import javax.servlet.annotation.WebServlet;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;
```

```
@WebServlet("/StudentServlet")
public class StudentServlet extends HttpServlet {
    @Override
    protected void doPost(HttpServletRequest request,
            HttpServletResponse response)
            throws ServletException, IOException {
        response.setContentType("text/html;charset=UTF-8");
        PrintWriter out = response.getWriter();
        try {
            int id = Integer.parseInt(request.getParameter("id"));
            String name = request.getParameter("name");
            String course = request.getParameter("course");
            String city = request.getParameter("city");

            Class.forName("com.mysql.cj.jdbc.Driver");
            String sql = "INSERT INTO student " +
                "(stud_id, stud_name, stud_course, stud_city) " +
                "VALUES (?, ?, ?, ?)";

            try (Connection con = DriverManager.getConnection(
                    "jdbc:mysql://localhost:3306/college",
                    "root", "");
                 PreparedStatement ps = con.prepareStatement(sql)) {
                ps.setInt(1, id);
                ps.setString(2, name);
                ps.setString(3, course);
                ps.setString(4, city);
                int result = ps.executeUpdate();
                if (result > 0) {
                    out.println("<h2>Student Inserted Successfully!</h2>");
                } else {
                    out.println("<h2>Insertion Failed!</h2>");
                }
            }
        } catch (Exception e) {
            throw new ServletException("Student insertion failed", e);
        }
    }
}
```

Note: Set your actual MySQL password instead of the empty string. Tomcat 9 uses javax.servlet; Tomcat 10+ requires jakarta.servlet imports and matching libraries.

Step 5 - Run and verify

Right-click project > Run As > Run on Server > Tomcat 9. Open **http://localhost:8080/StudentRegistration/index.jsp**. Fill the form and click Insert Student. Then execute:

```
USE college;
SELECT * FROM student;
```

Sample database row: 101 | Rahul | BCA | Jamnagar

Practical 2: Python Fibonacci Series

Aim: Print the Fibonacci series for a user-specified number of terms.

Step 1 - Create fibonacci.py

Open IDLE or VS Code and save a new file as **fibonacci.py**.

Step 2 - Copy-paste Python code

```
# Fibonacci Series Program
```

```
def fibonacci(n):
    a = 0
    b = 1
    print("Fibonacci Series:")
    for i in range(n):
        print(a, end=" ")
        c = a + b
        a = b
        b = c

print("----- Fibonacci Series -----")
n = int(input("Enter number of terms: "))
if n <= 0:
    print("Please enter a positive number.")
else:
    fibonacci(n)
print("\nProgram Completed.")
```

Step 3 - Run the program

In IDLE press F5, or in a terminal in the same folder run:

```
python fibonacci.py
```

Step 4 - Sample output

```
----- Fibonacci Series -----
Enter number of terms: 10
Fibonacci Series:
0 1 1 2 3 5 8 13 21 34
Program Completed.
```

Step 5 - Algorithm

1. Start. 2. Initialize a=0 and b=1. 3. Read n. 4. Repeat n times: print a, compute c=a+b, update a=b and b=c. 5. Stop.

Quick viva questions

JDBC: Java Database Connectivity. **PreparedStatement:** parameterized SQL statement. **executeUpdate():** executes INSERT/UPDATE/DELETE. **Fibonacci:** each term is the sum of the previous two.